NATIONAL MATHEMATICS CONTEST
PRIMARY-LEVEL PAPER 2.

NAMES: _____ DATE OF BIRTH: _____

SCHOOL/INSTITUTION: _____

CLASS: _____ DISTRICT: _____

INSTRUCTIONS TO CANDIDATES AND SUPERVISORS:

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PLEASE, NOTE CAREFULLY:

1. **WRITE YOUR ANSWERS IN THE SPACES PROVIDED. MORE SPACE IS AVAILABLE AT THE END OF THIS PAPER.**
2. Date: Saturday, 17th June, 2017.
3. Time: Exactly 2 hours and 15 Minutes (from 9 : 00am to 11 : 15am)
4. Type:
   (a) This paper consists of two sections and each section carries 50 marks.
   (b) Attempt as many questions as you can. The total marks scored from all the questions you attempt will be your final score.
   (c) No marks will be awarded for an answer if NO clear and logical layout of the working is shown.
5. All diagrams/figures in this contest are not to scale.
6. Immediately the paper is done, the answer script(s) must be forwarded to:

   David Ddumba Walakira,
   Contest Coordinator, Uganda Mathematical Society,
   c/o Department of Mathematics, Makerere University,
   P.O. Box 7062, Kampala.
   Telephone contact: $+256-414-540692$.

   Answer scripts should be received **latest, Monday** 20**th June,** 2017.
7. **ALL** *participants who qualified and sat Paper* 2 *are cordially invited to the certificate and/or prize giving ceremony on Saturday* 29*th July,* 2017 *at NOON at Makerere University, Kampala.*
8. National Mathematics Contest 2018 Paper 1 will be done on 17th March, 2018 at various centres.
9. **UGANDA MATHEMATICAL SOCIETY** wishes you success in this year's contest.

SECTION A : 5 marks each

Qn. 1. What numbers should be placed in places p and x to make the following statement true?

$$8 + \frac{p}{x} + \frac{3}{1000} = 8.403$$

Qn. 2. What integer is closest in value to $7 \times \frac{3}{4}$?

Qn. 3. What number gives a quotient of 6 and a remainder of 5 when divided by 8?

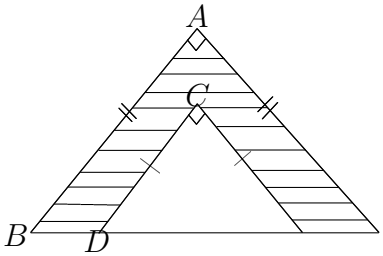
Qn. 4. A set of five different positive integers has an average of 11. What is the largest possible number in this set?

Qn. 5. In a basket ball shooting competition, each competitor shoots ten balls which are numbered $1, 2, 3, \ldots, 9, 10$. The number of points earned for each successful shot is equal to the number on the ball. If the competitor misses exactly two shots, which of the following scores is not possible and why?

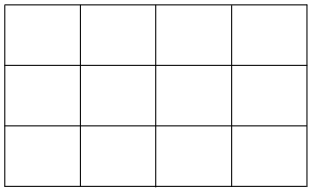
A) 52 *B*) 44 *C*) 41 *D*) 38 *E*) 35

Qn. 6. In the expression $\frac{a}{b} + \frac{c}{d} + \frac{e}{f}$, each letter is replaced by a different digit from $3, 4, 5, 7, 8, 9$. What is the largest possible value of this expression?

Qn. 7. Two isosceles right angled triangles are shown in the diagram with $AB = 5cm$ and $CD = 3cm$. What is the area of the shaded region?



Qn. 8. In the diagram, all the 12 small rectangles are the same size. Shade some of the rectangles until $\frac{1}{4}$ of $\frac{2}{3}$ of the diagram is shaded.

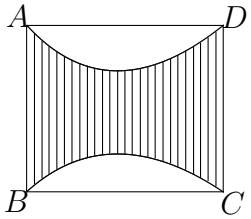


Qn. 9. Five students named Fred, Grace, Harriet, Iga, and Joan are seated around a circular table in that order. To decide who goes first in a game, they play countdown. Harriet starts by saying 204, with Iga saying 203. If they continue to count down in their circular order, who will eventually say 1?

Qn. 10. Paul, Queen, Rose, Sarah and Tom are sitting around a table. Queen sits in the chair between Paul and Sarah. Tom is not beside Sarah. Who is sitting between Queen and Rose?

SECTION B : 10 marks each

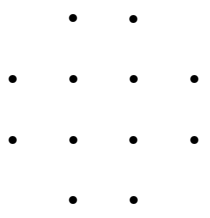
Qn. 11. $ABCD$ is a rectangle with two semicircles and $AD = 7cm$. The shaded area is $49cm^2$. (Take $\pi = \frac{22}{7}$).



(a) What is the area of the rectangle $ABCD$?

(b) What is the shortest distance between the semicircles?

Qn. 12. Twelve points are marked on a rectangular grid as shown. How many squares can be formed by joining four of these points?



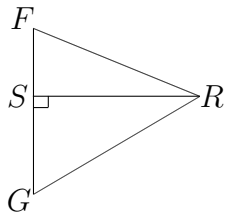
Qn. 13. A dishonest butcher priced his meat so that meat meant to be sold at Shs. $7,540$ was actually sold for Shs. $8,000$ per Kg. The butcher had sold 2200Kg of meat before being caught. He was fined Shs. $800,000$ plus 30% of the fine.

(a) How much extra profit did he dishonestly make?

(b) Did he remain with any profit after paying the fine?

Qn. 14. If $x * y = yx^2 + xy^2$, is $(-5 * 2) * 3 = 5 * (-2 * 3)$?

Qn. 15. The homes of Ferry (F), Sarah (S), Ruth (R) and Gideon (G) are marked as shown with straight lines joining them. Ferry is considering four routes to visit each of his friends:
1. $F \to R \to S \to G$; 2. $F \to S \to G \to R$; 3. $F \to R \to G \to S$; 4. $F \to S \to R \to G$.



If $FS = 5$Km, $SG = 9$Km, $SR = 12$Km, what is the difference between the longest and shortest trip if Ferry goes back to his home using the same route?

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SECTION A : 5 marks each

Qn. 1.

$$8.403 = 8 + \frac{4}{10} + \frac{0}{100} + \frac{3}{1000} \quad \Rightarrow \quad p = 4,\ x = 10$$

Qn. 2.

$$7 \times \frac{3}{4} = 5\frac{1}{4}$$

Nearer to 5 than 6. So the answer is 5.

Qn. 3.

$$(6 \times 8) + 5 = 48 + 5 = 53$$

Qn. 4. Sum of the 5 is $5 \times 11 = 55$. The smallest numbers are $1, 2, 3, 4$ to have the largest as

$$55 - (1 + 2 + 3 + 4) = 45$$

Qn. 5. If all the balls scored, total is

$$1 + 2 + 3 + \cdots + 10 = \frac{n}{2}(n + 1) = 55$$

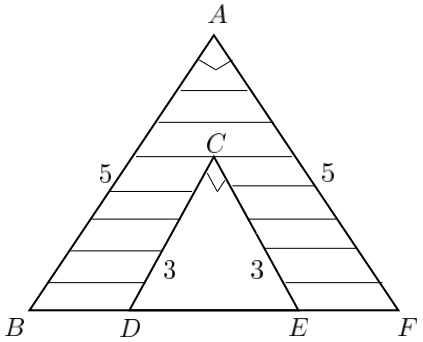
If 1 and 2 is missed, maximum is 52. If 9 and 10 is missed, maximum is 36. Therefore, every score between 36 and 52 is also possible, so 35 is NOT. Thus option E.

Qn. 6. Largest possible value is

$$\frac{9}{3} + \frac{8}{4} + \frac{7}{5} = 3 + 2 + 1.4 = 6.4$$

Qn. 7.

$$\begin{aligned} \text{Area of } ABF &= \frac{1}{2}(5)(5)\ cm^2 \\ \text{Area of } CDE &= \frac{1}{2}(3)(3)\ cm^2 \end{aligned}$$



Area of the shaded region is

$$\frac{1}{2}(5)(5) - \frac{1}{2}(3)(3) = 8 \text{ cm}^2$$

Qn. 8.

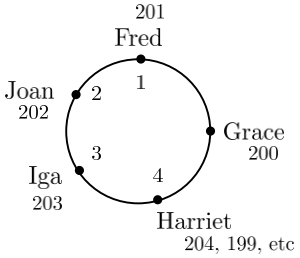
$$\frac{1}{4} \times \frac{2}{3} \times 12 = \frac{1}{6} \times 12 = 2$$

Shade any two squares.

Qn. 9. Harriet says 204 and will always say

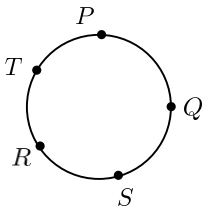
$$204 - 5n : 204, 199, \cdots, 4.$$

Harriet says 4, Iga 3, Joan 2, Fred 1.



Therefore it is Fred.

Qn. 10. Sarah since



SECTION B : 10 **marks each**

Qn. 11. (a) Total are of the $ABCD$ two semicircles

$$\pi r^2 = \frac{22}{7} \times \frac{7}{2} \times \frac{7}{2} = \frac{77}{2}\ cm^2 \ \text{ or } \ \frac{49}{4}\pi\ cm^2$$

Area of the rectangle is

$$\frac{77}{2} + 49 = \frac{175}{2}\ cm^2 \ \text{ or } \ 87\frac{1}{2}\ cm^2 \ \text{ or } \ 87.5\ cm^2$$

(b)

$$(AB)(AD) = AB(7) = 87.5 \quad \Rightarrow \quad AB = 12.5\ cm$$

Closest distance is $12.5 - (3.5 + 3.5) = 12.5 - 7 = 5.5\ cm$

Qn. 12. We have 5 small squares, 4 large squares and 2 huge squares. The total is 11 squares.

Qn. 13. (a) Profit is

$$(8,000 - 7,540) \times 2,200 = 460 \times 2,200 = 1,012,000$$

(b) The fine is

$$800,000 + (0.3)(800,000) = 1,040,000$$

The profit or loss is

$$1,012,000 - 1,040,000 = -28,000$$

No, he made a loss of $28,000$.

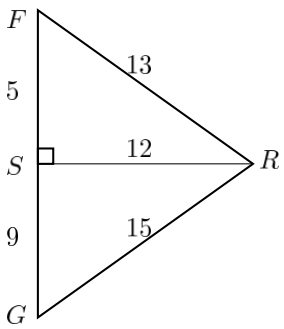
Qn. 14. If $x * y = yx^2 + xy^2$, is $(-5 * 2) * 3 = 5 * (-2 * 3)$?

$$\begin{aligned}
-5 * 2 &= (2)(-5)^2 + (-5)(2)^2 = 2 \times 25 - 5 \times 4 = 50 - 20 = 30 \\
(-5 * 2) * 3 &= (3)(30)^2 + (30)(3)^2 = 3 \times 900 - 30 \times 9 = 2700 + 210 = 2970 \\
-2 * 3 &= (3)(-2)^2 + (-2)(3)^2 = 3 \times 4 - 2 \times 9 = 12 - 8 = -6 \\
5 * (-2 * 3) &= (-6)(5)^2 + (5)(-6)^2 = -6 \times 25 + 5 \times 36 = -150 + 180 = 30
\end{aligned}$$

Therefore

$$(-5 * 2) * 3 = 2970 \neq 5 * (-2 * 3) = 30$$

Qn. 15. By Pythagoras theorem,



we have

$$FR = 13\ Km, \quad GR = 15\ Km$$

1	$F \to R \to S \to G = 34\ Km$
2	$F \to S \to G \to R = 29\ Km$
3	$F \to R \to G \to S = 37\ Km$
4	$F \to S \to R \to G = 32\ Km$

The shortest distance is

$$37 - 29 = 8\ Km$$